cut. Separable fat from all cuts was pooled to form raw and cooked composites. Separable fat included both external and seam fat in these composites. Separable lean was placed in a Cuisinart® food processor and homogenized for 35 seconds. Sample aliquots were frozen at -10°C until analysis.

Sample analyses. Proximate nutrients (moisture, total fat, ash, and protein) were determined on individual samples and composites of the separable fat. Raw and cooked samples of separable fat and the separable lean from the arm roast, bottom round steak, and top loin steak, trimmed to 1/8 inch external fat, were also analyzed for minerals (calcium, copper, iron, magnesium, manganese, phosphorus, potassium, selenium, sodium, and, zinc) and vitamins (niacin, thiamin, riboflavin, vitamins B_6, and B_{12}). Samples from the raw and cooked arm roast and separable fat were analyzed for vitamins A and E, total folate, and pantothenic acid. Raw samples from the arm roast were analyzed for amino acids. Data were released in SR16 (2003).

Grass-fed Beef

A collaborative study (Leheska, J.M. *et al*., 2008) was funded by the Beef Checkoff Program and conducted by America's Beef Producers, Texas Tech University, and USDA to determine the nutrient composition of grass-fed beef in the United States for inclusion in SR. The demand for grass-fed products has increased in recent years due to increased public interest in grass-fed production practices and nutrition. Crop variety, season, and geographic location can have an effect on the nutrient content of feedstuffs. In turn, the different types of feed given to cattle can affect weight gain, carcass characteristics, and nutrient content.

Sampling. Ground beef and strip steaks were collected on 3 separate occasions from 15 producers of grass-fed beef, representing 13 different states (Alabama, Arkansas, California, Colorado, Georgia, Idaho, Kentucky, Minnesota, Missouri, Montana, New Mexico, Texas, and Virginia). The sample collection protocol required that 2 steaks from 3 different animals be collected by each producer on each of the 3 separate occasions. The steaks were cut 2.54 cm thick from the 13th rib position of the strip loin. Similarly, 454 g of ground beef targeting 85% lean and 15% fat was collected by each producer from 3 different carcasses on each of 3 different occasions. When the specified lean to fat ratio (85/15) was not available they were asked to provide the next leanest ground beef (e.g., 88/12). The samples were then packaged appropriately and shipped frozen to Texas Tech University.

Sample preparations, grass-fed ground beef samples. After the ground beef samples had thawed properly they were frozen in liquid nitrogen and homogenized. Once homogeneity was reached aliquots of the samples were double bagged in labeled Whirl-Pak bags and stored at -80°C until subsequent analysis.

Sample preparations grass-fed strip steak samples: After proper thawing, the strip steak samples were weighed and dissected. The lean, fat, and refuse (connective tissue and scrap) of each steak was separated and weighed individually. Samples of cubed

strip steak were frozen in liquid nitrogen and homogenized using the same protocol as ground beef samples. Aliquots of the homogenized samples were double bagged in labeled Whirl-Pak bags and stored at -80°C until subsequent analyses.

Chemical Analysis. Analyses of proximate nutrients were performed at Texas Tech University. Following ether extraction, fat was determined in each sample using the Soxhlet method according to Official Method 991.36. Percent protein was determined by combustion using a LECO FP 2000 following AOAC Official Method 992.15. Percent moisture of the samples was analyzed by oven drying according to AOAC Official Method 8.2.1.1 and percent ash was determined by difference. Fatty acid analysis and cholesterol content was performed by a commercial laboratory using gas chromatography according to AOAC Official Methods 963.22 and 994.15. The University of North Carolina analyzed the grass-fed beef samples for choline by extracting choline compounds and quantifying by liquid chromatography-electrospray ionization-isotope dilution mass spectrometry. Total choline content of the samples was calculated as the sum of choline-contributing metabolites. Total fat, thiamin, vitamin B_{12}, and minerals (calcium, copper, iron, magnesium, manganese, phosphorus, potassium, selenium, sodium, and, zinc) were analyzed by a commercial laboratory using AOAC Official Methods. To validate all analytical procedures, quality control was monitored by insertion of certified reference materials and blind duplicates into the sampling course. Data on Grass-fed beef was released with SR21 (2008).

Ground Beef Products.

The USDA, in collaboration with America's Beef Producers and the University of Wisconsin, undertook a study funded by the Beef Checkoff Program to update the nutrient composition data for ground beef products in SR. None of the ground beef products contained extenders. According to Federal regulations, ground beef has no added water, phosphates, binders, or extenders, and shall not contain more than 30 percent fat (USDA, FSIS, Code of Federal Regulations). Ground beef is a unique meat product in that a wide range of formulations for this product are available in most US retail stores. In order to provide consumers and industry with the nutrient composition information for this variable product, the study was designed to establish the mathematical relationship between the various nutrients and the total fat content of raw ground beef through regression techniques. The ultimate aim was to use these relationships for predicting the nutrient composition for raw and prepared ground beef.

Sampling. For the first phase of this study, ground beef samples for each of three fat categories (label declarations of <12% fat, 12-22% fat, or >22% fat) were purchased from 24 retail outlets nationwide. In this sampling plan developed for the NFNAP (Pehrsson, P.R. *et al*., 2000), the country was divided into 4 regions, with 3 consolidated metropolitan statistical areas (CMSA) within each region, with 2 retail stores selected within each CMSA. To obtain updated data at lower levels of fat reflecting current retail market trends, a second phase of the study was conducted, using the NFNAP sampling plan with 12 nationwide retail locations to procure additional ground beef products of various fat levels.

Sample preparation. Ground beef products were analyzed in raw and cooked form. To achieve uniform sizing for broiled and pan-broiled patties, 112 g of ground beef were pressed into a patty mold. Patties were broiled in a preheated conventional oven for 8.7 min (final internal temperature of 160°F (71°C). Pan-broiled patties were broiled in a pre-heated Westbend® electric skillet for 11.75 min (final internal temperature of 160°F (71°C). Patties were cut in half to evaluate degree of doneness based on color. Ground beef crumbles were cooked in a pre-heated Westbend® electric skillet for 5.3 min (final internal temperature of 160°F (71°C)), and drained in a colander. The loaf was baked in a conventional oven at 325°F (163°C) for 41 min (final internal temperature of 160°F (71°C)). No fat was added during cooking. After cooking, all samples were stored at –24°C in sealed vacuum bags until homogenization and analysis.

Sample analyses. Proximate nutrients (moisture, total fat, ash, and protein) and cholesterol were determined on individual muscle samples from the chuck clod, bottom round, and the knuckle, both raw and cooked. Two composites composed of up to four samples each were analyzed for fatty acids, B vitamins (niacin, thiamin, riboflavin, vitamins B6 and B12), and minerals (calcium, magnesium, potassium, manganese, iron, phosphorus, sodium, copper, zinc and selenium) for each muscle group. A single nationally representative composite composed of three samples was used for analysis of choline, total folate, vitamins E and K for each muscle group. Cooking yields were also calculated based on initial (raw) and final cooked weights from all samples. These data were disseminated in SR18 (2005).

Nutrient analyses were conducted at either University laboratories or at a commercial testing laboratory using AOAC methods. Quality control measures included duplicate sampling and the use of control composites and NIST certified reference materials (SRM 1546: Meat Homogenate).

Statistics. Data were analyzed using mixed model regression analysis to obtain a regression equation for each nutrient and preparation method (SAS, 2004). Nutrient values from the first phase of the study were released in SR15 (2002) for ground beef products containing 5%, 10%, 15%, 20%, 25%, and 30% fat. The ground beef SR items include values for raw samples, broiled patties, pan-broiled patties, pan-browned crumbles, and baked loaf. The nutrient data from the first phase were combined with the proximate data from the second phase. Recent data from another beef study provided data for estimating values for retinol, vitamin E (α tocopherol), vitamin D and *trans*-fatty acid. Updated regression equations for each nutrient and for each preparation method were calculated. Nutrient values were then estimated from the equations for updating SR for 30, 25, 20, 15, 10, and 5% fat, and new profiles were created for 3% and 7% fat ground beef for each preparation method. The ground beef calculator, released on the NDL website in 2006 and updated in 2014, computes the nutrient profile for raw and prepared ground beef products at lean/fat levels between 97/3 and 70/30.

Beef Value Cuts

A new line of single–muscle roasts and steaks, fabricated from the outside round, the knuckle, and the chuck shoulder clod, were introduced to the retail market in 2001-2002. These cuts, the top blade steak (Infraspinatus), shoulder top and center steaks (Triceps brachii), shoulder tender (Teres major), tip center (Rectus femoris), tip side (Vastus lateralis), and bottom round (Biceps femoris), were tested for palatability and functionality. Furthermore, five of the six major cuts met the USDA definition of lean or extra-lean. USDA, in collaboration with America's Beef Producers and the University of Wisconsin, conducted a study funded by the Beef Checkoff Program to determine the nutrient profile of the Beef Value Cuts for inclusion in SR.

Sampling. Animal products were obtained from an IBP (Tyson) plant near Sioux City, Iowa. This plant draws cattle from a large number of feedlots and has nationwide product distribution. Twelve carcasses were identified by quality grade (high choice, average choice, and select) with yield grades of 2 or 3. Two carcasses were used for reserves and for training the meat cutting staff. There was sufficient product from 1 knuckle, 1 outside round, and 1 chuck clod to sample, prepare, and analyze five of the cuts. The Teres major is a very small muscle (~8 oz from 1 side) and would not provide a sufficient amount for all analyses. Therefore, one 15 pound box of choice (quality grade unknown) and one box of select Teres major muscles were purchased from the same plant. Removed beef value muscles were trimmed free of all external fat and heavy connective tissue. The denuded muscles were vacuum packaged and stored at -20°F until steak preparation.

Sample preparation. Muscles were cut into 1-inch thick steaks and weighed. Steaks were removed in pairs, one steak for raw analyses, the other to be cooked and analyzed in the cooked state. Steaks were cooked by grilling over a preheated portable gas grill. Steaks were turned when the internal temperature reached the midway point between the starting temperature and the final internal temperature (including post-cooking temperature rise) of 160°F (71°C) (medium degree of doneness). Steaks were placed on a wire rack for 3 min and then weighed to obtain the cooked weight. Raw and cooked steaks were stored at -20°F (-29°C) until time for nutrient analyses.

Sample analyses. Proximate nutrients (moisture, total fat, ash, and protein) and cholesterol were determined on individual muscle samples from the chuck clod, bottom round, and the knuckle, both raw and cooked. Composites of three samples from each of these muscle groups were pooled into composites and analyzed for fatty acid content. Individual samples from the knuckle muscles were also analyzed for of minerals (calcium, copper, iron, magnesium, manganese, phosphorus, potassium, selenium, sodium, and, zinc) and vitamins (niacin, riboflavin, thiamin, vitamins B_6 and B_{12}). Samples from the raw and cooked knuckle muscles were also analyzed for vitamins A and E. No vitamins or minerals were analyzed on samples from the chuck clod or bottom round; NDL imputed these values based on nutrient values from the arm roast and bottom round. Cooking yields calculations were based on initial (raw) and final cooked weights from all samples. These data were disseminated in SR18 (2005).

Beef Nutrient Database Improvement Study:

A collaborative research study was undertaken by NDL with scientists at the National Cattlemen's Beef Association (NCBA), Colorado State University (CSU), Texas A & M University (TAMU), and Texas Tech University (TTU) to update nutrient information in the USDA National Nutrient Database for Standard Reference (SR). This entailed updating the food and nutrient composition for beef cuts currently in SR, and adding new cuts, which had been introduced in the market place. The first phase of this study involved cuts from the chuck: Brisket, Mock Tender Steaks, Top Blade Steaks, Shoulder Steaks Boneless, Shoulder Clod Roasts, Boneless Chuck Short Ribs, Denver Steaks, Chuck Eye Steaks, Country Style Ribs, America's Beef Roast, Underblade Steaks and Roasts, and Beef for Stewing. Most of these cuts are new with the exception of the Shoulder Steaks which replaced the older Clod Steak data (NDB#s 23533, 13943, 23536, 13946, 23554, 23516). The second phase of this study involved cuts from the rib and plate: Back Ribs, Rib Eye Roast, Rib Eye Steak, Outside Skirt, and Inside Skirt. During the second phase of the Beef Nutrient Database Improvement Study, a separate study on Beef Alternative Merchandising (BAM) beef cuts was also conducted. BAM cuts were developed by the beef industry to utilize all the potential meat from today's larger subprimals and traditional subprimals, and to respond to customers' desire for leaner, more health-conscious portions. BAM cuts are leaner and smaller than more traditional cuts. The boneless beef cuts added to SR from the BAM study were: Ribeye Filet, Ribeye Petite Roast, Ribeye Cap Steak, Top Loin Filet, Top Loin Petite Roast, Top Sirloin Filet, Top Sirloin Cap Steak, and Top Sirloin Petite Roast. In the third phase of this study, focusing on the loin and the round, these cuts were added to SR: Top Loin Steak trimmed to 0" fat, Top Loin Steak trimmed to 1/8" fat, T Bone Steak, Porterhouse Steak, Tenderloin Steak, Tenderloin Roast, Top Round Steak, Top Round Roast, Eye of Round Steak, and Eye of Round Roast. For each cut in this study, nutrient values are provided in SR in both raw and cooked forms for "separable lean only" and "separable lean and fat", and for quality grades Select, Choice, and "all grades". Data from this project were incorporated into SR and were also disseminated in a separate report on the NDL web site titled "USDA Nutrient Dataset for Beef Retail Cuts", revised periodically, with version 3.0 released in 2013.

Sampling: Beef carcasses for the study were selected from six different major packing plants, representing the different regions of the US Each university was assigned two different packing plants. The sampling plan was developed for 36 animals. In order to get true retention and yield data, an A and a B side of the animal carcass was needed; thus the total animal count came to 72. When selecting the carcasses certain properties were considered as part of the sampling plan protocol: quality grade (upper choice, lower choice, select), yield grade (YG2, YG3), gender (steer or heifer), and genetics (dairy or non-dairy). Each university was responsible for identifying and obtaining beef chucks that fit into the sampling matrix. The universities assessed and recorded carcass data at the packing plants, properly identified each selected cut and shipped the product back to their respective meat laboratories. Products were fabricated into the needed retail cuts for this study within 14-21 days postmortem.

Retail cuts were properly identified and vacuum packaged and held frozen until cooking or dissection. The retail product was cooked according to protocols developed for each cut. Cooked and raw products were dissected; weights for each component (separable lean, separable fat, and refuse) were obtained. Total weights of raw and cooked (prior to and after cooking) cuts were obtained. Samples were then homogenized and composited.

The compositing plan was developed to establish an effective and efficient statistical design for nutrient analyses of the beef cuts. The plan consisted of 4 different compositing levels: an animal level (36 animals) where all the samples were analyzed; a six composite level; a three composite level; and a national composite level. This was done for both raw and cooked samples. Different nutrients were analyzed at each composite level.

Sample preparation: The various beef cuts were analyzed in raw and cooked form. The following cooking methods were used: grilling, roasting, and oven-braising. Frozen raw samples were tempered under refrigeration (0-4°C) for 24 to 48 hours based on the appropriate size and weight of the cut. The appropriate temperatures and weights were recorded prior to cooking. The thermocouple was placed in the geometric center or thickest portion of the meat piece. The probe positioning did not affect the product's contact with the cooking surface. For small or thin beef cuts, the thermocouple was used periodically to check the internal temperature of samples throughout the cooking process.

Cooking Procedures:

Grilling - The grill was preheated to 195°C (383°F). The beef samples were evenly spaced in the center of cooking grate. The grill lid was closed and the sample was cooked to an internal temperature of 70°C (158°F). Tongs or spatulas were used to remove samples from the grill. Beef samples were allowed to stand while monitoring the internal temperature rise until temperatures began to decline. The point right before the temperature declines (highest temperature reached) was considered the final internal temperature of the cooked sample. Beef samples were then chilled uncovered in the refrigerator (2-4° C) for 24 ± 1 hour before dissection.

Roasting - The oven was preheated to 160°C (325°F). The beef sample(s) were positioned in the center of the rack in the roasting pan, no oil or water was added, and the pan was not covered. The roasting pan with the beef sample was positioned on the oven rack in center of oven and roasted to an internal temperature of 60°C (140°F). The beef samples were removed from the oven. The thermocouple probe remained in place and samples were allowed to stand while monitoring the internal temperature rise until temperatures began to decline. The point right before the temperature declines (highest temperature reached) was considered the final internal temperature of the cooked sample. The beef samples were then chilled uncovered in refrigeration (2-4° C) for 24 ± 1 hour before dissection.

Oven-Braising - The beef samples were placed in a preheated pan and were

"browned/seared", turning as needed for even browning on all sides. The pan drippings were poured off and the volume (mL) of drippings was measured. The thermocouple was then applied in the geometric center or thickest portion of the meat piece. A small amount of distilled, deionized water was added until the water reached one third-the thickness of the meat. The liquid was held at a simmer, the pan was covered with a lid, and placed in the Dutch oven. The Dutch oven was then placed in a preheated 120°C (250°F) oven. The beef samples simmered and cooked until an internal temperature of 85°C was reached. The samples were removed from the oven keeping the thermocouple probe in place and were allowed to stand while monitoring the internal temperature rise until temperatures began to decline. The point right before the temperature declines (highest temperature reached) was considered the final internal temperature of the cooked sample. The beef sample(s) were removed from the cooking liquid and the cooking liquid yield and volume were documented. The beef samples were then chilled uncovered in the refrigerator (2-4° C) for 24 ± 1 hour before dissection. In phase two, the back ribs were oven-braised and the "browned/seared" step was not performed.

Nutrient Analysis: At the animal level, only proximates were analyzed. At the next level, the six composite level, the following nutrients were analyzed: Proximates (fat, moisture, protein, and ash), fatty acids including long-chain fatty acids and CLAs, total cholesterol, minerals (Ca, Fe, Mg, P, K, Na, Zn, Cu and Mn), selenium, vitamin E, vitamin D, and B vitamins including B12, B6, riboflavin, and niacin. At the 3 composite level, amino acids and retinol were analyzed. At the final National composite level, total choline and the other B vitamins (thiamin and pantothenic acid) were analyzed. The pooled fat samples, both raw and cooked, from all the cuts were analyzed for all nutrients.

The techniques for analyzing the proximate nutrients are as follows: Protein by combustion, total fat by extraction and acid hydrolysis, ash by gravimetric, and moisture by forced air. The minerals calcium, magnesium, iron, zinc, copper, and manganese were analyzed by atomic absorption spectroscopy (AAS). Potassium and sodium were analyzed by emission spectrometry, and selenium by hydride generation. Retinol, vitamin E, and vitamin D were analyzed by high-performance liquid chromatography (HPLC) methods. Choline was analyzed by liquid chromatography-electrospray ionization-isotope dilution mass spectrometry (LC/ESI/IDMS). B-vitamins such as thiamin and riboflavin were analyzed by fluorometric methods. Niacin, pantothenic acid, vitamin B6, and vitamin B12 were analyzed by microbiological methods. Amino acids such as tryptophan were analyzed by alkaline hydrolysis-HPLC, cystine and methionine by performic oxidation-HPLC, and all other amino acids by acid hydrolysis-HPLC. Hydroxyproline was analyzed using a colorimetric method, cholesterol by a gas chromatographic (GC)/direct saponification method not using derivatization, and fatty acids by gas-liquid chromatography(GLC).

New Zealand Beef Study

A study was conducted in collaboration with the New Zealand Meat Industry Association to determine nutrient composition of 32 imported New Zealand beef cuts and offal items (10 samples per cut) which are available in retail grocery stores in the US. This study was done along with study of New Zealand lamb cuts, described in the Notes on Lamb section of this report. The beef cuts were bolar blade, brisket naval end, brisket point end, chuck eye roll, cube roll, eye round, flank, flat, hind shin, inside cap-off, oyster blade, ribs, rump center, strip loin, tenderloin, heart, tongue, tripe, kidney, knuckle and liver. These cuts were selected by the members of the New Zealand Meat Industry Association with different items being supplied from different meat plants throughout New Zealand. Retail cuts fabricated from the meat carcasses were prepared for dissection, homogenization and nutrient analysis at Massey University, New Zealand. Weights for component factors such as separable lean, intramuscular and subcutaneous fat (separable fat), bone and connective tissue were determined. Nutrient values were added to SR for each of these cuts in both raw and cooked form (using a braised, fast roasted, fast fried, or boiled method) for “separable lean only” and “separable lean and fat“ items.

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Breakfast Cereals (Food Group 08)

Introduction

Food Group 08 foods are identified as breakfast cereals to clearly distinguish them from cereal products used mainly as ingredients or typically consumed at meals other than breakfast (see Food Group 20 Cereal Grains and Pasta). The Breakfast Cereals group of more than 360 items includes two major categories: ready-to-eat (RTE) and to-be-cooked (hot) cereals. The majority of breakfast cereals are listed by brand name. The number and level of fortification nutrients differ appreciably between breakfast cereals, resulting in many unique products that can't adequately be described generically. The majority of major brand breakfast cereals are included in SR, accounting for over 80% of the retail market.

Breakfast cereals generally consist of one or more cereal grains, either as whole grains or milled portions, as a major constituent. The continuum of grain content goes from less than 50% for some presweetened RTE cereals and approaches 100% for hot cereals. The predominant grains for RTE cereals are corn, wheat, oats and rice. Additional ingredients such as sweeteners, flavoring or texturizing macroingredients (including fruit, nuts, and oil), microingredient flavors or colors, and nutritional fortificants and shelf life preservatives may be added (Caldwell, 2000). Manufacturing processes generally used for RTE cereals include: flaked, extruded flakes, gun-puffed whole grains, extruded gun-puffed, oven-puffed, shredded whole grains, and extruded shredded methods.

Fortification: Addition of vitamins and/or minerals to grain products began in the late 1930's with selected nutrients (primarily thiamin, riboflavin, niacin, iron, and calcium) being added in amounts to restore the natural content of the grain which may have been modified during processing (enrichment). A standard of identity, effective in 1942, established standards for unenriched and enriched farina. Enrichment standards were developed for corn grits in 1947 (Park, 2001; FDA 2012). In 1955, nutrients were first added to breakfast cereals in amounts higher than those of the whole grain itself. By 1969 many RTE cereals were fortified with 25% of the US RDA for thiamin, riboflavin, niacin, vitamin B6, vitamin B12, A, C, and folic acid; with iron at 10 to 25% RDA; and some cereals with vitamin D at 10% RDA (Steele, 1976). A recognition of the importance of folate in prevention of neural tube defects led to an FDA regulation, effective in 1998, requiring folate fortification of specific flours and grains (see Table of Standards of Enrichment, in Notes on Foods for Cereal Grains and Pasta), including enriched farina, which is used as a hot cereal (Phillips, 2010; Rader, 2000). Ready-to-eat cereals were not affected by the regulation. Today, nearly all processed ready-to-eat breakfast cereals are fortified with vitamins and/or minerals at varying levels.

Addition of nutrients presents technological problems – some vitamins are not heat stable; others are affected by pressure; and some can produce undesirable tastes and odors (Steele, 1976). Incorporation of fortificants before processing provides uniform